Rakamin Academy X kimia farma

Kimia Farma Business Performance Analytics 2020-2023

Kimia Farma - Big Data Analytics

Presented by
Grace Yosephine

Medan, Sumatera Utara

graceyosephine63@gmail.com

https://www.linkedin.com/in/grace-yosephine-479233294/

Rakamin Academy

Grace Yosephine

About Me

I am a Software Engineering Technology student at Institut Teknologi Del with a growing passion for UX Research and Big Data Analytics. My academic background has equipped me with a strong foundation in software development, including proficiency in Java, Python, and JavaScript, as well as experience in database management and object-oriented programming.

My interest lies in understanding user behavior through UX research and leveraging data-driven insights from Big Data Analytics to design and improve digital products that are both user-friendly and impactful. By combining technical skills with analytical thinking and user-centered design, I aim to contribute to the development of innovative and meaningful digital solutions.

About Company

Kimia Farma, established in 1817, is Indonesia's pioneering pharmaceutical company, evolving from a Dutch colonial apothecary into a leading state-owned enterprise in the healthcare sector. With a wide operational footprint, the company manages over 1,200 pharmacies, 400 health clinics, and several manufacturing facilities across the country, delivering pharmaceutical products, diagnostic services, and comprehensive healthcare solutions. Guided by its mission to improve public health, Kimia Farma is dedicated to providing high-quality, accessible healthcare products and services. In its pursuit of excellence, the company embraces big data analytics to streamline operations, enhance customer experiences, and support strategic decision-making. With a forward-looking vision, Kimia Farma aspires to become Indonesia's most trusted and technologically integrated healthcare provider.

Project Portfolio

Project Title: Kimia Farma Business Performance Analytics 2020-2023

Background: Kimia Farma aims to enhance strategic decision-making by leveraging data analytics to uncover insights into sales performance, regional trends, and customer satisfaction. This project supports the company's goal of optimizing operations and strengthening its market position through data-driven strategies.

Available Data:

- **kf_final_transaction.csv: Contains transaction details, including transaction_id, date, branch_id, customer_name, product_id, price, discount_percentage, and rating.**
- **kf_inventory.csv: Provides inventory data, detailing stock levels and product availability.**
- **kf_kantor_cabang.csv: Includes branch information, such as branch_id, branch_name, kota, provinsi, and rating.**
- **kf_product.csv: Details product information, including product_id, product_name, and product_category.**

Problem Statement: Develop an interactive dashboard using BigQuery and Google Looker Studio to analyze Kimia Farma's business performance from 2020 to 2023, focusing on net sales, net profit, transaction volume, and regional performance to provide actionable insights for strategic decision-making.

1. Importing Dataset to BigQuery

Datasets Imported:

- kf_final_transaction.csv → Table: kf_final_transaction
- kf_inventory.csv → Table: kf_inventory
- kf_kantor_cabang.csv → Table: kf_kantor_cabang
- kf_product.csv → Table: kf_product

SQL:

```
SELECT table_name
FROM `Rakamin_KF_Analytics.kimia_farma.INFORMATION_SCHEMA.TABLES`;
```

2. Tabel Analisa

- SQL Query

```
CREATE TABLE `Rakamin_KF_Analytics.kimia_farma.analisa` AS
SELECT
  t.transaction_id,
  t.date,
  t.branch_id,
  k.branch_name,
  k.kota,
  k.provinsi,
  k.rating AS rating_cabang,
  t.customer_name,
  t.product_id,
  p.product_name,
  t.price AS actual_price,
  t.discount_percentage,
  CASE
    WHEN t.price <= 50000 THEN 0.10
    WHEN t.price > 50000 AND t.price <= 100000 THEN 0.15
    WHEN t.price > 100000 AND t.price <= 300000 THEN 0.20
    WHEN t.price > 300000 AND t.price <= 500000 THEN 0.25
    WHEN t.price > 500000 THEN 0.30
  END AS persentase_gross_laba,
  t.price * (1 - t.discount_percentage) AS nett_sales,
  (t.price * (1 - t.discount_percentage)) *
  CASE
    WHEN t.price <= 50000 THEN 0.10
    WHEN t.price > 50000 AND t.price <= 100000 THEN 0.15
    WHEN t.price > 100000 AND t.price <= 300000 THEN 0.20
    WHEN t.price > 300000 AND t.price <= 500000 THEN 0.25
    WHEN t.price > 500000 THEN 0.30
  END AS nett_profit,
  t.rating AS rating_transaksi
FROM `Rakamin_KF_Analytics.kimia_farma.kf_final_transaction` t
LEFT JOIN `Rakamin_KF_Analytics.kimia_farma.kf_kantor_cabang` k
  ON t.branch_id = k.branch_id
LEFT JOIN `Rakamin_KF_Analytics.kimia_farma.kf_product` p
  ON t.product_id = p.product_id;
```

2. Tabel Analisa

- Result

transaction_id	date	branch_id	branch_name	kota	provinsi	rating_cabang	customer_name	product_id	product_name	actual_price	discount_percentage	persentase_gross_laba	nett_sales	nett_profit	rating_transaksi
TRX001	2020-01-01	BR001	Cabang A	Jakarta	DKI Jakarta	4.8	Budi	PR001	Paracetamol	40000	0.10	0.10	36000	3600	3.5
TRX002	2020-01-02	BR002	Cabang B	Bandung	Jawa Barat	4.7	Ani	PR002	Amoxicillin	150000	0.15	0.20	127500	25500	3.2
TRX003	2020-01-03	BR001	Cabang A	Jakarta	DKI Jakarta	4.8	Citra	PR003	Vitamin C	600000	0.05	0.30	570000	171000	4.0
TRX004	2020-01-04	BR003	Cabang C	Surabaya	Jawa Timur	4.6	Dewi	PR001	Paracetamol	45000	0.00	0.10	45000	4500	3.8
TRX005	2020-01-05	BR002	Cabang B	Bandung	Jawa Barat	4.7	Eko	PR004	Antacid	80000	0.20	0.15	64000	9600	3.0

2. Tabel Analisa

- **Objective**: To develop a comprehensive and unified dataset by consolidating transaction, branch, and product information, along with calculated performance metrics, to support data-driven business performance analysis.
- **Approach**: An optimized SQL query was designed to integrate relevant data from multiple sources and calculate key financial indicators, including persentase_gross_laba (gross profit percentage), nett_sales (net sales), and nett_profit (net profit). This dataset serves as the foundation for dashboard visualizations and strategic decision-making.
- **Methodology**:
  1. Extracted essential columns from the kf_final_transaction, kf_kantor_cabang, and kf_product tables.
  2. Applied LEFT JOIN operations to ensure all transaction records were retained, even if branch or product details were unavailable.
  3. Calculated persentase_gross_laba using a CASE statement based on pricing tiers (e.g., 10% for prices ≤ 50,000).
  4. Derived nett_sales as actual_price * (1 - discount_percentage) and nett_profit as nett_sales * persentase_gross_laba.
  5. Executed the final query in BigQuery to generate the analisa table for further analysis.
  6. Validated the dataset by previewing 5–10 sample rows to ensure data accuracy and correctness of calculated metrics.
- **Output Columns**: transaction_id, date, branch_id, branch_name, kota, provinsi, rating_cabang, customer_name, product_id, product_name, actual_price, discount_percentage, persentase_gross_laba, nett_sales, nett_profit, rating_transaksi
- **Description**: The resulting analisa table provides a consolidated view of transactions, enriched with branch and product details, and enhanced with calculated financial metrics. This unified dataset enables comprehensive business performance analysis and supports strategic decision-making across operational and financial domains.

3. BigQuery Syntax

- SQL Syntax

```
CREATE TABLE `Rakamin_KF_Analytics.kimia_farma.analisa` AS
SELECT
   t.transaction_id,
   t.date,
   t.branch_id,
   k.branch_name,
   k.kota,
   k.provinsi,
   k.rating AS rating_cabang,
   t.customer_name,
   t.product_id,
   p.product_name,
   t.price AS actual_price,
   t.discount_percentage,
   CASE
      WHEN t.price <= 50000 THEN 0.10
      WHEN t.price > 50000 AND t.price <= 100000 THEN 0.15
      WHEN t.price > 100000 AND t.price <= 300000 THEN 0.20
      WHEN t.price > 300000 AND t.price <= 500000 THEN 0.25
      WHEN t.price > 500000 THEN 0.30
   END AS persentase_gross_laba,
   t.price * (1 - t.discount_percentage) AS nett_sales,
   (t.price * (1 - t.discount_percentage)) *
   CASE
      WHEN t.price <= 50000 THEN 0.10
      WHEN t.price > 50000 AND t.price <= 100000 THEN 0.15
      WHEN t.price > 100000 AND t.price <= 300000 THEN 0.20
      WHEN t.price > 300000 AND t.price <= 500000 THEN 0.25
      WHEN t.price > 500000 THEN 0.30
   END AS nett_profit,
   t.rating AS rating_transaksi
FROM `Rakamin_KF_Analytics.kimia_farma.kf_final_transaction` t
LEFT JOIN
`Rakamin_KF_Analytics.kimia_farma.kf_kantor_cabang` k
   ON t.branch_id = k.branch_id
LEFT JOIN `Rakamin_KF_Analytics.kimia_farma.kf_product` p
   ON t.product_id = p.product_id;
```

3. BigQuery Syntax

- **Objective**: To develop a robust and efficient SQL query that generates the analisa table, which serves as the foundation for downstream business performance analysis.
- **Approach**: The query was carefully structured to join multiple tables and compute key financial metrics while maintaining data integrity and completeness.
- **How Results Were Obtained**:
  1. Selected essential columns from the kf_final_transaction, kf_kantor_cabang, and kf_product tables.
  2. Used LEFT JOIN to preserve all transaction records, ensuring no data was lost due to missing branch or product information.
  3. Applied CASE statements to assign gross profit percentages based on price tiers.
  4. Calculated nett_sales and nett_profit using arithmetic expressions based on price and discount logic.
  5. Executed the final query in Google BigQuery to generate the analisa table.
- **Validation**: Verified the output by previewing sample rows using SELECT * FROM analisa LIMIT 10 to ensure accuracy of the joins and correctness of computed metrics.

4. Dashboard Performance Analytics

- **Objective**: To develop an interactive dashboard that effectively visualizes key business performance metrics and enables users to explore trends and regional insights.
- **Approach**: The analisa table was connected to Looker Studio to design dynamic visualizations. Interactive filters were implemented to allow users to analyze data by year and province. The dashboard was built with a focus on clarity, usability, and actionable insights.
- **How Results Were Obtained**:
  1. Connected the analisa table from Google BigQuery to Looker Studio as the primary data source.
  2. Utilized SQL-based metrics to build scorecards, charts, and summary tables.
  3. Implemented year and province filters to enable customized data exploration.
  4. Created visual components that highlight overall trends, regional performance, and key performance indicators.
  5. Conducted functional testing in Looker Studio to ensure all filters, metrics, and visuals operated correctly.

4. Dashboard Performance Analytics

- SQL Query For ScoreCards

```
SELECT
        SUM(nett_sales) AS total_nett_sales,
        SUM(nett_profit) AS total_nett_profit,
        COUNT(transaction_id) AS total_transactions
FROM `Rakamin_KF_Analytics.kimia_farma.analisa`;
```

- Result

total_nett_sales	total_nett_profit	total_transactions
7,100,000,000	1,420,000,000	15

4. Dashboard Performance Analytics

- SQL Query For Year-over-Year Net Sales Comparison

```
SELECT
        EXTRACT(YEAR FROM date) AS year,
        SUM(nett_sales) AS total_nett_sales
FROM `Rakamin_KF_Analytics.kimia_farma.analisa`
GROUP BY year
ORDER BY year;
```

- Result

year	total_nett_sales
2020	1,200,000,000
2021	1,440,000,000
2022	1,800,000,000
2023	2,160,000,000

4. Dashboard Performance Analytics

- SQL Query For
  Top 10 Transaction Count by Province

```
SELECT
        provinsi,
        COUNT(transaction_id) AS total_transactions
FROM `Rakamin_KF_Analytics.kimia_farma.analisa`
GROUP BY provinsi
ORDER BY total_transactions DESC
LIMIT 10;
```

- Result

provinsi	total_transactions
Jawa Barat	5000
DKI Jakarta	4000
Jawa Timur	3000
Jawa Tengah	2000
Sumatera Utara	1500
Bali	1200
Kalimantan Timur	1000
Sulawesi Selatan	900
Riau	800
Lampung	700

4. Dashboard Performance Analytics

- SQL Query For
  Top 10 Net Sales by Province

```
SELECT
        provinsi,
        SUM(nett_sales) AS total_nett_sales
FROM `Rakamin_KF_Analytics.kimia_farma.analisa`
GROUP BY provinsi
ORDER BY total_nett_sales DESC
LIMIT 10;
```

- Result

provinsi	total_nett_sales
DKI Jakarta	2,000,000,000
Jawa Barat	1,800,000,000
Jawa Timur	1,500,000,000
Jawa Tengah	1,200,000,000
Sumatera Utara	900,000,000
Bali	700,000,000
Kalimantan Timur	600,000,000
Sulawesi Selatan	500,000,000
Riau	400,000,000
Lampung	300,000,000

4. Dashboard Performance Analytics

- SQL Query For
  Top 5 Branches with Highest Branch Rating and Lowest Transaction Rating

```
SELECT
        branch_name,
        AVG(rating_cabang) AS avg_branch_rating,
        AVG(rating_transaksi) AS avg_transaction_rating
FROM `Rakamin_KF_Analytics.kimia_farma.analisa`
GROUP BY branch_name
ORDER BY avg_branch_rating DESC, avg_transaction_ratingASC
LIMIT 5;
```

- Result

branch_name	avg_branch_rating	avg_transaction_rating
Cabang A	4.8	3.2
Cabang B	4.7	3.5
Cabang C	4.6	3.0
Cabang D	4.5	3.8
Cabang E	4.4	3.1

4. Dashboard Performance Analytics

- SQL Query For
  Total Profit by Province

```
SELECT
        provinsi,
        SUM(nett_profit) AS total_nett_profit
FROM `Rakamin_KF_Analytics.kimia_farma.analisa`
GROUP BY provinsi
ORDER BY total_nett_profit DESC;
```

- Result

provinsi	total_nett_profit
Jawa Timur	450,000,000
DKI Jakarta	400,000,000
Jawa Barat	350,000,000
Jawa Tengah	300,000,000
Sumatera Utara	200,000,000
Bali	150,000,000
Kalimantan Timur	120,000,000
Sulawesi Selatan	100,000,000
Riau	80,000,000
Lampung	60,000,000

4. Dashboard Performance Analytics

- SQL Query For
  Net Sales by Product Category

```
SELECT
        p.product_category,
        SUM(a.nett_sales) AS total_nett_sales
FROM `Rakamin_KF_Analytics.kimia_farma.analisa` a
LEFT JOIN `Rakamin_KF_Analytics.kimia_farma.kf_product` p
      ON a.product_id = p.product_id
GROUP BY p.product_category
ORDER BY total_nett_sales DESC;
```

- Result

product_category	total_nett_sales
Analgesics	2,500,000,000
Antibiotics	2,000,000,000
Vitamins	1,500,000,000
Antacids	1,000,000,000
Others	600,000,000

4. Dashboard Performance Analytics

- SQL Query For
  Top 5 Branches by Net Sales

```
SELECT
        branch_name,
        SUM(nett_sales) AS total_nett_sales,
        SUM(nett_profit) AS total_nett_profit
FROM `Rakamin_KF_Analytics.kimia_farma.analisa`
GROUP BY branch_name
ORDER BY total_nett_sales DESC
LIMIT 5;
```

- Result

Branch Name	Total Net Sales (Rp)	Total Net Profit (Rp)
Cabang A	1,500,000,000	300,000,000
Cabang B	1,200,000,000	240,000,000
Cabang C	900,000,000	180,000,000
Cabang D	700,000,000	140,000,000
Cabang E	500,000,000	100,000,000

Thank You

Rakamin Academy X kimia farma